



Mayank Jha
Computer Science & Engineering
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M.Tech.
Gender: Male
DOB: 23/09/2002

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2027	null
Graduation	IIIT Delhi	IIIT Delhi	2024	7.11
Graduation Specialization: Computer Science and Biosciences				

M.TECH PROJECT AND SEMINAR

Game Theoretic Approaches for Security Planning

July 2026 - Present

Prof. Swaprava Nath, Dept. of CSE, M.Tech Project

IIT Bombay

- Studied game-theoretic decision-making in defence through a 28-paper review and 3 frameworks, exploring minimax, Counterfactual Regret Minimization, and Online Learning to understand how defenders adapt against an adaptive adversary in SAM-SEAD engagements
- Modeled a 22-aircraft, 10-barrier patrol problem as a resource-allocation optimization game, deriving optimal coverage through a 3-stage LP and mixed-strategy formulation under a 1,210-hour/month flying constraint and exposing a $2 \times$ capacity gap for guaranteed full coverage
- Converted the optimal allocation into a concrete patrol schedule using LP duality and column generation to construct an integer mixed strategy, randomized phase offsets to generate unpredictable patrol timetables; proved $\leq K + 1$ iteration termination and worst-case optimality

PROJECTS

Impact of Label Noise on ML Generalization

July 2025 – Nov 2025

Prof. Abir De, Dept. of CSE, Course Project

IIT Bombay

- Implemented Generalized Cross-Entropy and confident-learning label cleaning pipelines, recovering up to 16.3pp and 13.9pp of noise-induced accuracy loss respectively through controlled same-architecture experiments
- Analyzed hyperparameter sensitivity under noise, finding SVM regularization shift $10 \times$ and Random Forest's 10-vs-200 tree gap widen from 1.6pp to 15.1pp at 33% noise; tested cross-domain transfer on Fashion-MNIST and reported the non-replicating result
- Engineered a reproducible ML benchmarking pipeline with CLI-driven experiments, Docker, GitHub Actions CI, and pytest, evaluating 6 classifiers across 4 noise mechanisms and 5 noise levels over 5 seeds and 750 experiment runs

Search & Retrieval Engine

July 2026 – Present

Prof. Soumen Chakrabarti, Dept. of CSE, Course Project

IIT Bombay

- Engineered an end-to-end search engine spanning a concurrent Python crawler, C++ indexing engine, and FastAPI query service; implemented custom tokenization and a flat-array positional inverted index with LEB128-encoded positions and from-scratch multi-term BM25 scoring across 20K documents, 8M+ postings, and 850K+ unique terms
- Built a hybrid semantic retrieval pipeline integrating 384-dimensional sentence-transformer embeddings with BM25 and PageRank through min-max normalized weighted fusion, improving real-corpus nDCG@10 from 0.46 to 0.63 while automatically identifying and quantifying a $\sim 58 \times$ PageRank selection bias in relevance judgments
- Designed a byte-exact cross-language binary index format with LEB128 delta encoding and versioned sections, preserving compatibility across an $81 \times$ corpus expansion with zero downstream code changes; containerized the pipeline and calibrated 2–5GB memory limits, establishing a stable 5GB bound with ~ 730 MB headroom under real query load

Multi-Model Content Moderation & NLP Platform

July 2023 – Nov 2023

Self Project

- Engineered an NLP classification pipeline over 7.7M chat messages, evaluating 8 classical models, a recurrent model, and DistilBERT on an identical 200K balanced split; achieved a 22.5% relative F1 gain over the best classical baseline
- Developed a 13-class fallacy analysis engine with structure-aware preprocessing, LIME, multi-seed evaluation, and 1,000-resample bootstrap testing; achieved a $6 \times$ gain in paraphrase robustness, reducing prediction flips from 13.7% to 2.3%
- Architected a multi-model FastAPI serving layer with per-model failure isolation, graceful fallback, model discovery, and health endpoints across two task-specific APIs; containerized with Docker and validated via end-to-end HTTP execution

NETRA: Full-Stack News Aggregation & Personalization Platform

July 2025 – Nov 2025

Prof. Om Damani, Dept. of CSE, Course Project

IIT Bombay

- Engineered a full-stack news platform with Flask, React and PostgreSQL/pgvector, integrating JWT authentication, relational integrity and 127 backend tests at 89% coverage; deployed end-to-end across Supabase, Render and Vercel
- Built a 384-dimensional semantic recommendation pipeline using sentence-transformer embeddings with cosine similarity, recency decay, category affinity and engagement signals; evaluated ranking quality with nDCG@5 of 0.241 on a 2,027-article corpus
- Optimized data retrieval through bulk SQL aggregation, reducing per-page database queries from 80 to 2, while implementing an $O(nk)$ greedy story-clustering algorithm with 0.35s runtime and debugging multi-worker scheduler failures

PostgreSQL Index Advisor

July 2026 – Present

Prof. S. Sudarshan, Dept. of CSE, Course Project

IIT Bombay

- Engineered a Java 21/Spring Boot index advisor that screens candidates with **HypoPG** and validates winners using real **CREATE INDEX + EXPLAIN ANALYZE**; live production testing measured up to **96.5%** query improvement
- Diagnosed a **HikariCP concurrency race** and cache-warming measurement bias through concurrent-load testing and symmetric median-of-3 benchmarking; secured dynamic DDL with schema allowlisting and quoted identifiers
- Implemented greedy **marginal-utility index-set selection** with B-tree prefix-aware domination pruning, eliminating redundant indexes across workloads; HypoPG screen-then-validate benchmarking delivered **~1.2×** average speedup across 5 workloads

Distributed Genomic Streaming Pipeline

Jan 2024 – July 2024

Self Project

- **Found and fixed a hidden flaw** in a similarity-matching system built from scratch using Bloom filters and MinHash for scanning DNA sequencing data; discovered the similarity check breaks down when comparing a short DNA fragment against an entire genome and replaced it with an exact-match technique, **cutting Bloom candidates by 96% (6,274 → 244)** across 1.35M real sequencing reads
- **Trained a 19-feature PyTorch neural network** to classify flagged DNA sequences, testing against realistic sequencing errors and a wide range of simulated infection levels; achieved a **97.6–100% detection rate with 0 false positives**
- **Built a fault-tolerant streaming pipeline with Apache Kafka** to ingest DNA sequencing data across 8 parallel partitions with crash-safe PostgreSQL persistence; proved **12s recovery with zero message loss** across 20K reads under container-kill recovery tests in Docker

Distributed Key-Value Cache

Jan 2024 – July 2024

Self Project

- **Engineered a distributed in-memory key-value cache from scratch**, implementing a TCP protocol, fixed-thread-pool server, LRU eviction, and TTL expiration; a 3-node cluster sustained **6,000/6,000 successful operations at 20 concurrent clients**, versus 2,149/6,000 (36%) on 1 node
- **Diagnosed the load collapse to TCP ephemeral-port exhaustion**, using `netstat` to identify 16,340 `TIME_WAIT` sockets; benchmarked 6 load scenarios, measuring **1.7×** higher throughput with 3-node partitioning at 10 clients
- **Hardened correctness by closing 3 race conditions** in capacity checks, TTL updates, and LRU state, using `ReentrantLock` to protect `LinkedHashMap`'s mutating-on-read access order; validated with **26 unit, protocol, and integration tests**

TECHNICAL SKILLS

Languages, Tools and Frameworks: Python, C++, Java, SQL, JavaScript, React, TypeScript, Redux, Vite, Tailwind

Machine Learning: PyTorch, scikit-learn, Deep Learning, NLP, Embeddings, CNNs, Classical ML, Model Evaluation

Algorithms & Data Structures: Data Structures, Algorithms, Dynamic Programming, Graph Algorithms, Greedy Algorithms, String Algorithms, Optimization, Searching, BM25, PageRank, Suffix Arrays, Bloom Filters

Backend & Databases: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MySQL, Redis, pgvector, Query Optimization, Indexing

Systems & Engineering: Multithreading, Concurrency, Apache Kafka, Client-Server Architecture, TCP Sockets, Fault Tolerance, Git, Linux, Docker, Debugging, Performance Optimization

COMPETITIVE PROGRAMMING

CodeChef Rating: 1450+ , Codeforces Rating: 1100+ , LeetCode Contest Rating: 1500+ & 270+ Problems Solved

POSITIONS OF RESPONSIBILITY

Teaching Assistant, CS101: Computer Programming and Utilization, IIT Bombay

Aug 2025 – Nov 2025

Personally Guided 6 students through C++ fundamentals via doubt-clearing sessions resulting in improved course performance.

Teaching Assistant, CS218: Design and Analysis of Algorithms, IIT Bombay

Jan 2026 – May 2026

Prepared 4 quizzes, designed rubrics, supervised examinations, and graded 100+ evaluation papers to support course assessment.

Teaching Assistant, CS6001: Game Theory & Algorithmic Mechanism Design, IIT Bombay

July 2026 – Present

RELEVANT COURSEWORK

IIT Bombay: Software Engineering, Indexing and Retrieving Text and Graphs*(ongoing course), Foundations of Machine Learning

IIIT Delhi: Operating Systems, Data Structures and Algorithms, Advanced Programming, Analysis and Design of Algorithms, Fundamentals of Database Management Systems, Artificial Intelligence, Natural Language Processing, Game Theory

EXTRACURRICULARS

Community Work with E-Vidyaloka , Sports & Fitness: Recreational Sports and Regular Strength Training

HOBBIES AND INTERESTS

Playing Chess, Reading, Learning Philosophy & Psychology, Exploring Scriptures, Spirituality & Self-Reflection, Exploring Game Theory, International Relations, Economics & History, Building Tools to Solve Everyday Problems, Cinephile, Music Enthusiast